Olist E-Commerce — Weekly Business Review

Execution Timestamp: **13 October 2026 17:04** | Observation Period: **Sep 2016 – Aug 2018** | Total Orders: **99,441** | Delivered Orders: **96,478**

1. Executive Summary & Dataset Overview

This Weekly Business Review evaluates operational performance for the Olist marketplace across **99,441 total orders** in 27 Brazilian states from September 2016 through August 2018 (with overall order activity extending through October 2018). Of these, **96,478 orders (97.0%)** were recorded as delivered. Among the **96,470 delivered orders with valid delivery and estimated timestamps**, the marketplace North Star metric—**On-Time Delivery Rate**—stands at **91.89%** (88,644 on-time deliveries), with a directly calculated **Late-Delivery Rate of 8.11%** (7,826 late deliveries). The fleetwide **Average Review Score is 4.09 / 5.00**, and the customer **Overall Repeat-Purchase Rate is 3.06%**.

2. Modular ELT Pipeline Architecture

The automated pipeline follows a reliable four-tier ELT architecture: (1) **Validation & Atomic Ingestion**: validates 9 source CSV files and expected column schemas; stages all tables into temporary tables before executing a single atomic transactional swap to guarantee zero partial-state reloads. (2) **Source Data Quality Audit**: evaluates raw tables for nulls, duplicates, orphaned keys, and anomalous amounts before transformation, halting immediately if mandatory checks fail. (3) **Staging & Mart Transformations**: executes typed DDL in staging and builds an order-level analytics mart (analytics.orders_mart) with exactly 1 row per order using CTE pre-aggregation to prevent join fan-out. (4) **SQL Analytics & Automated Reporting**: executes pure PostgreSQL analytical queries, generates 4 Matplotlib charts, and compiles this PDF.

3. Marketplace Metrics Framework

Metric Name	Formula & Eligible Population	Observed Value	Strategic Purpose & Business Context
North Star: On-Time Delivery Rate	Delivered <= estimated / 96,470 eligible orders * 100	**91.89%**	Primary customer trust benchmark; measured on delivered orders with both timestamps non-null.
Average Review Score	AVG(review_score) across reviewed orders in mart	**4.09 / 5.0**	Direct reflection of customer satisfaction and fulfillment quality across all reviewed orders.
Late-Delivery Rate	Delivered > estimated / 96,470 eligible orders * 100	**8.11%**	Direct measure of logistics SLA breach on the exact same eligible delivered population.
Order Cancellation Rate	Canceled orders (625) / 99,441 total orders * 100	**0.63%**	Quantifies checkout friction, inventory stockouts, and pre-fulfillment buyer abandonment.
Overall Repeat-Purchase Rate	Unique customers with >= 2 eligible orders / 95,560 total unique customers * 100	**3.06%**	Measures buyer retention using persistent customer_unique_id across non-canceled orders.

Data Quality Audit & Integrity Report

Source Layer Audit Executed: **13 October 2026 17:04** | Ledger Table: analytics.dq_audit

1. Data Quality Check Results (Raw Table Evaluation)

Source data quality checks run against raw tables prior to staging and mart transformations. Mandatory checks (HARD FAIL) abort the pipeline if defects are detected, preventing corrupted data from entering analytics.

Check File	Description	Failing Rows	Severity	Audit Status
dq_01_order_id	Null or duplicate order IDs in raw orders	0	HARD FAIL	PASS
dq_02_delivery_dates	Delivered timestamp earlier than purchase timestamp	0	SOFT EXCEPTION	PASS
dq_03_payment_reconciliation	Payment total vs item+freight reconciliation (>0.05 BRL)	260	SOFT EXCEPTION	FAIL_SOFT
dq_04_customer_integrity	Orders with missing customer foreign key references	0	HARD FAIL	PASS
dq_05_review_scores	Review scores outside valid 1-5 range or null	0	SOFT EXCEPTION	PASS
dq_06_order_item_integrity	Orphaned order items or non-positive item prices	0	SOFT EXCEPTION	PASS

2. Empirical Payment Reconciliation Analysis (DQ-03)

Executing sql/dq_03_payment_reconciliation.sql at its documented threshold of **> R$0.05** identifies exactly **260 orders** with payment discrepancies (a tighter threshold of > R$0.01 identifies 303 orders). Grouping discrepancies by payment method reveals the empirical distribution below:

Payment Method(s)	Order Count	Avg Discrepancy	Min Discrepancy	Max Discrepancy
credit_card	246	+11.81 BRL	-51.62 BRL	+182.81 BRL
debit_card	7	-7.42 BRL	-16.50 BRL	-1.60 BRL
credit_card, voucher	7	+2.50 BRL	+0.36 BRL	+9.26 BRL

3. Data Quality Policies & Anomaly Treatment

- **DQ-01 (Null / Duplicate Order IDs)**: *Hard Failure*. An order_id must uniquely identify each transaction. 0 duplicate or null order IDs were found across all 99,441 raw order rows.
- **DQ-02 (Delivery Timestamp Precedence)**: *Soft Exception*. Validates delivery timestamp >= purchase timestamp. 0 inverted delivery records were detected.
- **DQ-03 (Payment Reconciliation Policy)**: *Documented Exception*. Of the 260 discrepancies > R$0.05, 246 involve credit card payments where total payments exceed or fall short of item+freight totals (average diff +11.81 BRL), with 7 debit card orders and 7 combined credit card/voucher orders. Because source tables do not itemize external fees or discounts, records are retained intact without artificial adjustment.
- **DQ-04 (Customer Referential Integrity)**: *Hard Failure*. Ensures every raw order maps to a valid customer in raw_customers. 0 orphaned orders detected.
- **DQ-05 (Review Score Range Validation)**: *Soft Verification*. Confirms review scores are strictly integers between 1 and 5. 0 invalid scores found in raw reviews.
- **DQ-06 (Order Item Referential & Price Integrity)**: *Soft Verification*. Confirms line items reference valid orders and prices are positive. 0 orphaned or negative price rows detected.

SQL Analytics & Deep-Dive Findings

Direct PostgreSQL analytical query execution across orders, categories, and repeat buyer behaviors

1. Query 1: Month-over-Month Revenue & Growth Trend (Recent Sample Months)

Month	Delivered Revenue (BRL)	Prior Month Revenue	Change (BRL)	MoM Growth %
2017-11	R$ 1,153,364	R$ 751,117	R$ 402,247	53.6%
2017-12	R$ 843,078	R$ 1,153,364	R$ -310,286	-26.9%
2018-01	R$ 1,077,887	R$ 843,078	R$ 234,809	27.9%
2018-02	R$ 966,168	R$ 1,077,887	R$ -111,719	-10.4%
2018-03	R$ 1,120,598	R$ 966,168	R$ 154,430	16.0%
2018-04	R$ 1,132,879	R$ 1,120,598	R$ 12,281	1.1%
2018-05	R$ 1,128,775	R$ 1,132,879	R$ -4,104	-0.4%
2018-06	R$ 1,011,978	R$ 1,128,775	R$ -116,796	-10.4%
2018-07	R$ 1,027,807	R$ 1,011,978	R$ 15,829	1.6%
2018-08	R$ 985,492	R$ 1,027,807	R$ -42,316	-4.1%

2. Query 2: Top 3 Product Categories by Revenue in Key States

State	Top Product Category (English Translation)	Delivered Revenue (BRL)	State Rank
MG	Health Beauty	R$ 175,008	1
MG	Bed Bath Table	R$ 156,483	2
MG	Watches Gifts	R$ 132,312	3
RJ	Watches Gifts	R$ 188,265	1
RJ	Bed Bath Table	R$ 175,648	2
RJ	Health Beauty	R$ 159,005	3
RS	Bed Bath Table	R$ 72,428	1
RS	Furniture Decor	R$ 66,414	2
RS	Computers Accessories	R$ 61,844	3
SP	Bed Bath Table	R$ 549,062	1
SP	Health Beauty	R$ 510,255	2
SP	Watches Gifts	R$ 448,583	3

3. Query 4: Overall Repeat-Purchase Rate

Using persistent customer_unique_id identifiers across **95,560 unique buyers** with non-canceled orders, we identify **2,923 repeat buyers** (>=2 orders), resulting in an organic **Overall Repeat-Purchase Rate of 3.06%**. This indicates a predominantly single-transaction buyer base, highlighting significant growth potential via lifecycle retention programs.

Marketplace Performance Visualizations

Analytical visualizations generated directly from live SQL outputs

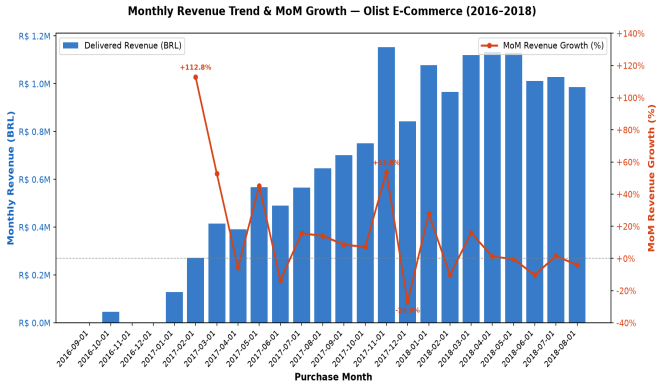


Figure 1: Monthly Revenue & MoM Growth

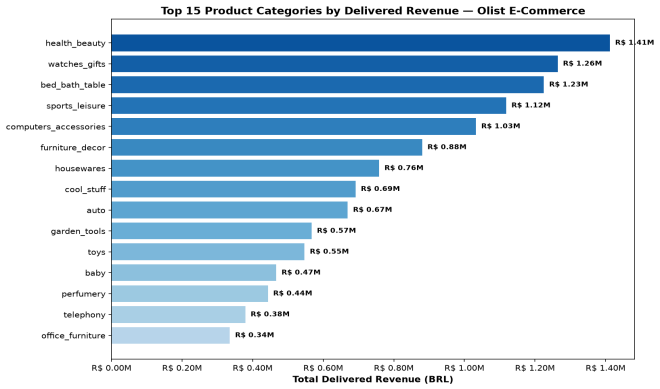


Figure 2: Top 15 Product Categories by Revenue

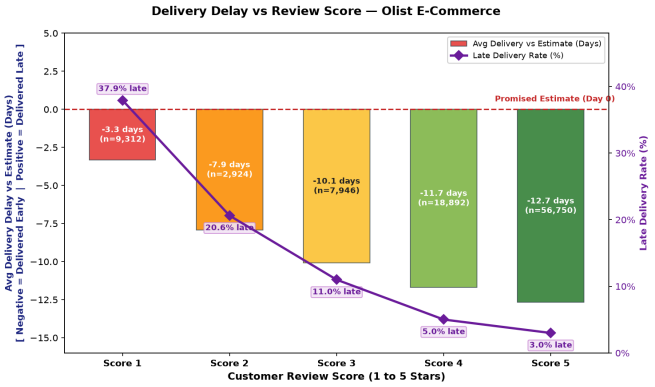


Figure 3: Delivery Delay vs Review Score

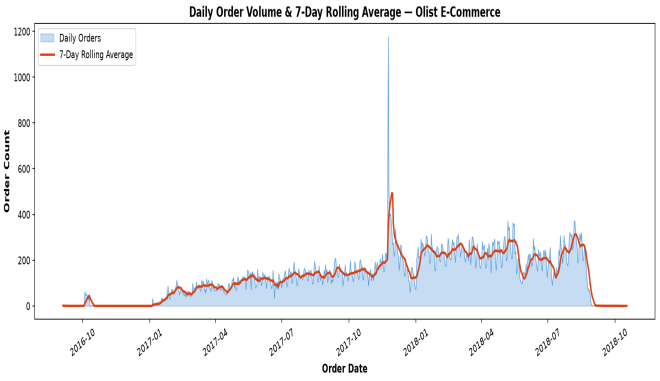


Figure 4: Daily Order Volume & 7-Day Rolling Avg

EXECUTIVE MEMORANDUM

TO: Vice President of Marketplace Operations | **DATE:** 13 October 2026 | **FROM:** Lead Data Analytics Team

SUBJECT: Empirical Correlation Between Delivery Delay and Customer Review Scores — Action Plan

1. Executive Finding

Observational analysis across **95,824 delivered orders with valid customer reviews** (out of 96,470 eligible deliveries) demonstrates a strong empirical association between delivery timeliness and review ratings. While orders receiving 5-star reviews (n=56,750) were delivered on average **12.68 days before their estimated delivery date** (with only a 3.0% late rate), orders receiving 1-star reviews (n=9,312) averaged **3.32 days earlier than estimated** (-3.32 days vs estimate), while **37.9% were still delivered late (12.6x higher than 5-star orders)**. These data reflect observational associations and do not prove that delivery timing is the sole causal driver of review scores.

2. Empirical Evidence (Query 5 Output)

Review Score	Delivered Orders (n)	Avg Delay vs. Estimate (Days)	Late Orders (%)
Score 1 / 5	9,312	-3.32 days	37.9%
Score 2 / 5	2,924	-7.93 days	20.6%
Score 3 / 5	7,946	-10.08 days	11.0%
Score 4 / 5	18,892	-11.68 days	5.0%
Score 5 / 5	56,750	-12.68 days	3.0%

3. Why This Finding Matters

With our overall repeat-purchase rate currently at **3.06%**, buyer retention represents our greatest organic growth lever. Highly rated transactions (5-star reviews comprise 59.2% of reviewed deliveries) correlate with early arrival. Conversely, the **7,826 late shipments (8.11% of all 96,470 eligible deliveries)** correlate with disproportionate negative reviews, eroding buyer trust and generating customer support overhead.

4. Proportionate Operational Recommendations

Based on observed correlations, we recommend targeted logistics interventions:
• *Dynamic Buffer Protocol*: Introduce a calibrated safety buffer to estimated delivery dates on routes with high delay variance to set realistic customer expectations.
• *Carrier & Route Review*: Benchmark carrier transit performance by regional corridor and establish SLA tracking for postal handoffs.
• *Seller Dispatch Monitoring*: Implement alerts for fulfillment partners to encourage dispatch within promised handling windows.

5. Measurable Success Criterion

Increase the marketplace **On-Time Delivery Rate from 91.89% to >=95.0%** (reducing late orders from 7,826 to <=4,823) and lift fleetwide **Average Review Score from 4.09 to >=4.30 / 5.00** within **120 days** of protocol rollout.

6. Material Caveats & Methodological Limitations

- **Correlation vs. Causation**: While delivery delays correlate heavily with negative reviews, product defects, seller communication, and packaging also contribute.
- **Historical Observation Period**: Delivered commercial operations span Sep 2016 – Aug 2018 (with non-delivered records extending to Oct 2018); current postal network and carrier infrastructure must be accounted for in baseline calibration.
- **Payment Reconciliation**: At the > R$0.05 threshold, exactly 260 orders show discrepancies (dominated by 246 credit card transactions differing by an average of +11.81 BRL), retained as documented exceptions.